KU LEUVEN

ARENBERG DOCTORAL SCHOOL
Faculty of Engineering Technology

Dynamic reality modeling for AECO applications

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Dissertation presented in partial fulfillment
of the requirements for the degree of
Doctor of Engineering Technology (PhD)

October 2026

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Uitgegeven in eigen beheer, Jelle Vermandere, Gent (Belgium)

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List of Abbreviations

AECO Architecture, Engineering, Construction and Operations. 1

BIM Building Information Model. 1, 12

CSDF Coloured Signed Distance Field — an extension of the SDF representation in which each voxel additionally stores per-voxel colour information, enabling joint geometry and appearance manipulation. 12

DRM Dynamic Reality Model. 2, 3, 5, 6, 9, 12–15, 18, 19

iMMS Indoor Mobile Mapping System. 24

NeRF Neural Radiance Field — a learned continuous function mapping 3D coordinates and viewing directions to colour and volume density, enabling novel view synthesis. 12

SDF Signed Distance Function. 3, 12

SLAM Simultaneous Localisation and Mapping. 24

TLS Terrestrial Laser Scanner. 24

ToF Time-of-Flight. 25

UDF Unsigned Distance Field — a scalar field encoding the absolute (unsigned) distance from any point in space to the nearest surface, applicable to open or non-watertight geometry. 12

UV UV coordinates — two-dimensional texture coordinates assigned to mesh vertices that define the mapping between the 3D surface and a 2D texture image. 10

XR Extended Reality. 3, 10, 13, 24

Chapter 1

Introduction

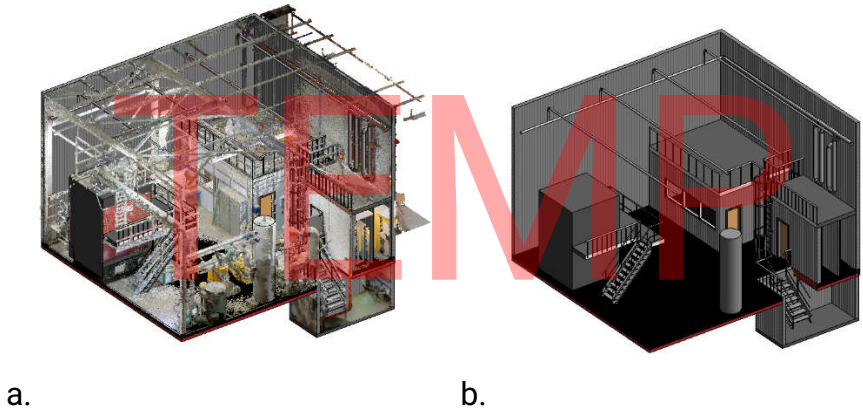
1.1 General context

The Architecture, Engineering, Construction and Operations (AECO) industry is undergoing a rapid digital transformation, driven by advances in reality capture technologies such as laser scanning and photogrammetry [7]. These technologies enable the efficient acquisition of high-resolution representations of the physical environment, forming a foundation for data-driven workflows and digital twins [16].

Two complementary paradigms currently dominate the representation of the built environment. On the one hand, reality capture-based models provide highly detailed and visually accurate reconstructions of real-world scenes, making them well suited for documentation, inspection and visualization [6]. On the other hand, Building Information Modelling (BIM) represents environments as structured collections of semantic, parametric objects, enabling efficient querying, modification and simulation [5].

Each paradigm offers distinct advantages, but also exhibits inherent limitations. Reality capture models preserve geometric and visual fidelity, yet remain static, incomplete and unstructured, lacking explicit object decomposition and semantic relationships [22]. In contrast, BIM models provide a dynamic and structured representation of buildings, but often lack geometric detail and visual realism, as they are typically created during the design phase and do not fully reflect as-built conditions [5]. (Figure 1.1)

This contrast highlights a fundamental limitation in current digital modelling



TEMP

Figure 1.1: Comparison between reality capture and Building Information Modelling approaches.

workflows: no single representation simultaneously provides realism, completeness and interactivity. Bridging this gap requires combining the strengths of both paradigms into a unified representation.

To address this challenge, this research introduces a novel representation, referred to as a *Dynamic Reality Model* (DRM). The DRM represents the environment as a collection of individual, manipulable objects, while preserving the geometric and visual fidelity of the captured scene. In doing so, it combines the structured and dynamic nature of BIM with the highly accurate and visually detailed reality capture data.

The resulting representation enables a wide range of applications that require both accuracy and flexibility. In renovation and refurbishment workflows, designers can evaluate design alternatives directly within the as-is environment, supporting tasks such as layout optimisation and feasibility analysis without requiring full manual reconstruction [42]. In facility management and reorganisation scenarios, objects such as furniture or equipment can be repositioned, replaced or removed, enabling efficient space planning while maintaining consistency with real-world constraints [16]. Furthermore, the combination of visual fidelity and object-level interactivity enables the use of real-world environments in interactive applications such as simulation, training and gaming [1].

Despite this potential, a fundamental barrier remains in transforming reality

capture data into such dynamic representations. This barrier stems from the combined effects of occlusions and the unstructured nature of the captured data. Scan-based representations only capture visible surfaces from limited viewpoints, leaving significant portions of the environment unobserved due to occlusions caused by clutter, other objects or restricted sensor access [44]. At the same time, the data lacks an explicit structural organisation, representing the scene as a dense collection of points or surfaces without clear object boundaries [22].

While recent learning-based approaches attempt to address the issue of missing geometry using implicit representations such as SDFs [29] or occupancy fields [13], and scene-level methods infer unseen regions [18], these approaches primarily focus on geometric completion. They do not address the combined challenge of reconstructing missing data while simultaneously introducing the structured, object-level representation required for interaction.

In addition to these structural challenges, built environments are inherently dynamic, continuously evolving over time due to usage, modifications and maintenance activities. As a result, any DRM that aims to accurately represent the physical environment must be updated regularly to remain relevant.

Traditionally, such updates rely on high-end sensing technologies, such as terrestrial laser scanning, which provide accurate and detailed observations but are time-consuming and costly to acquire [6]. This limits the frequency at which complete datasets can be captured. In contrast, recent advances in mobile and XR-based capture devices enable fast and accessible data acquisition, allowing smaller updates to be recorded more frequently, albeit at lower accuracy and completeness [42, 41].

This creates an opportunity to complement high-quality baseline datasets with incremental updates from lower-end devices. However, it also introduces new challenges, as the resulting data is heterogeneous in terms of accuracy, density and coverage. Integrating these observations into a consistent and up-to-date representation requires careful handling of temporal differences, data quality and spatial alignment [7, 41].

As a result, despite significant advances in data acquisition and reconstruction, there is currently no unified framework that transforms partial, heterogeneous scan data into complete, structured and dynamic scene representations. Existing methods address individual aspects of this problem, but fail to integrate them into a coherent and scalable pipeline.

1.2 Problem statement

Transforming partially scanned indoor environments into complete, structured and dynamic representations remains an open challenge in current 3D reconstruction workflows. This challenge is inherently multi-stage, requiring the integration of heterogeneous data, the recovery of missing geometry and the transformation of static reconstructions into interactive environments.

A first challenge lies in the alignment and updating of heterogeneous data. Different sensing technologies produce data with varying levels of accuracy, density and noise, resulting in inconsistent representations of the same environment [22]. Aligning these data sources into a common reference frame is particularly difficult in indoor settings, where global positioning systems are unavailable [42]. In addition, environments evolve over time, requiring new partial observations to be integrated into existing datasets. Efficiently updating scenes without reprocessing the entire dataset remains a significant challenge, requiring robust methods for both registration and incremental integration [41].

A second major challenge is the incompleteness of reality capture data. Scan-based devices record only what is visible from their measurement positions, leaving occluded regions unobserved [46]. This affects both the background scene and individual objects, producing data with missing geometry at multiple scales [43]. Effective completion additionally requires the reliable separation of individual objects from their surroundings, which is particularly difficult in cluttered environments where object boundaries are ambiguous and partially hidden [44].

A deeper difficulty is that scene-level and object-level completion are typically treated as separate problems, each with limitations that the other cannot resolve. Scene-level completion methods [24, 11, 47, 19, 20, 8, 36, 30] treat the environment as a monolithic volume and infer missing geometry globally. While this recovers regions outside the scanner's field of view, it cannot resolve occlusions between objects that are in contact: two chairs pushed together, a cabinet against a wall, or stacked equipment all present hidden surfaces that a global scene model cannot distinguish from empty space. Object-level completion methods [29, 13, 35, 12], by contrast, operate on objects already isolated from the scene, discarding the very environmental context — surrounding walls, floors and neighbouring objects — that could inform the plausible extent and boundaries of the missing geometry. Resolving this mutual dependency requires a framework in which scene and object completion are coupled through shared geometric constraints rather than treated in isolation.

Even when complete geometric reconstructions of both the scene and individual

objects are available, transforming them into a dynamic and interactive environment remains a non-trivial problem. This requires their analysis and structuring to support higher-level interactions [2]. Objects must be represented in a way that enables operations such as repositioning, replacement or modification, which goes beyond purely geometric reconstruction [**?**].

A final challenge lies in the evaluation of reconstruction and completion methods. Reliable ground-truth data for occluded regions is rarely available in real-world datasets, making quantitative evaluation difficult [17, 9]. The absence of ground truth for hidden surfaces means that existing benchmarks cannot directly assess the quality of completed geometry in the regions that matter most. Synthetic datasets offer an alternative, but must accurately replicate real-world scanning conditions — including noise, occlusion patterns and sensor-specific characteristics — to produce evaluation results that transfer to practice [21, **?**].

In summary, transforming partial scan data into dynamic scene representations requires addressing challenges in data alignment and updating, geometry completion at both scene and object level, and scene dynamification. These challenges are interdependent and must be solved within a unified and structured framework.

1.3 Research aim

The primary objective of this research is to develop a modular and adaptable methodology for transforming partially scanned indoor environments into complete, structured and dynamic representations, referred to as DRMs. This objective addresses the lack of a unified pipeline identified in the previous section, where current approaches treat alignment, completion and interaction as separate problems rather than components of a coherent workflow.

Instead of relying on a monolithic end-to-end solution, the proposed methodology is structured as a sequence of interconnected modules. Each module performs a specific task, producing an output that serves as input for the next stage. This modular design enables flexibility in handling heterogeneous and incomplete data, while allowing intermediate validation and refinement when required.

The pipeline consists of the following stages, as outlined in Figure 1.2:

- **Scene alignment and updating**
- **Scene segmentation and object isolation**
- **Scene completion**

- **Object completion**
- **Scene dynamification**

The conceptual role of each stage and the terminology associated with it are defined in Chapter 2, which establishes the representational and methodological foundation for the pipeline.

A key contribution of this work is the development of a DRM, which combines the geometric and visual fidelity of reality capture data with the structured and interactive nature of object-based models. This enables objects to be manipulated, repositioned and analysed within a representation that remains consistent with the physical environment.

To support evaluation, this research introduces a synthetic dataset generation framework that produces partial scans paired with complete ground-truth geometry. Because real-world datasets cannot provide ground truth for occluded regions — the very regions that completion methods must reconstruct — such a synthetic dataset is essential for quantitative assessment under controlled conditions that reflect real-world scanning characteristics.

The scope of this research is limited to single-room indoor environments containing furniture or industrial objects. This constraint enables a focused investigation of densely furnished scenes, where occlusions, object interactions and data heterogeneity are significant but manageable.

In summary, this research establishes a modular pipeline that integrates alignment, segmentation, completion and dynamification into a coherent workflow. By combining the strengths of reality capture and structured representations, the proposed methodology enables the creation of dynamic models from partial scan data.

1.4 Thesis Outline

This thesis follows the logical progression from data acquisition to the generation of dynamic scene representations. Each chapter builds upon the previous one forming a coherent pipeline from raw scan data to DRMs.

Chapter 2 introduces the background and terminology related to scene dynamification and 3D representations. It defines key concepts such as dynamic scenes, object-level representations and completion, positions the proposed DRM within the broader context of existing modelling paradigms, and explains the conceptual role of each pipeline stage.

Chapter 3 focuses on data acquisition, covering both real-world scanning and synthetic dataset generation. It discusses the characteristics and limitations of different sensing technologies, including occlusions, noise and coverage, and introduces the V-Scan synthetic dataset designed to enable controlled evaluation of reconstruction methods.

Chapter 4 addresses the alignment and updating of multi-source and multi-temporal data. It presents methods for integrating heterogeneous datasets into a consistent representation and explores strategies for incrementally updating scenes as new observations become available.

Chapter 5 focuses on object detection and occlusion analysis. It introduces methods for decomposing unstructured scan data into individual objects and for estimating occluded regions at both object and scene level, providing the basis for subsequent completion steps.

Chapter 6 presents the scene completion methodology, focusing on the reconstruction of structural elements of the environment. It describes how incomplete observations are converted into a consistent geometric and textured representation of the scene.

Chapter 7 explores object completion techniques, comparing different approaches for reconstructing partially observed objects. The chapter evaluates their performance and discusses their integration within the overall pipeline.

Chapter 8 introduces the dynamification of completed scenes. It describes how static reconstructions are transformed into structured, interactive representations, enabling object-level manipulation and interaction within the environment.

Chapter 9 concludes the thesis by summarizing the main contributions and discussing the limitations of the proposed methodology, as well as directions for future research.

Chapter 10 explores the valorization potential of the research, highlighting applications in areas such as renovation, facility management and interactive environments, and discussing opportunities for real-world deployment.

Through this structured approach, the thesis reflects the full transformation pipeline from partial scan data to dynamic and interactive scene representations.

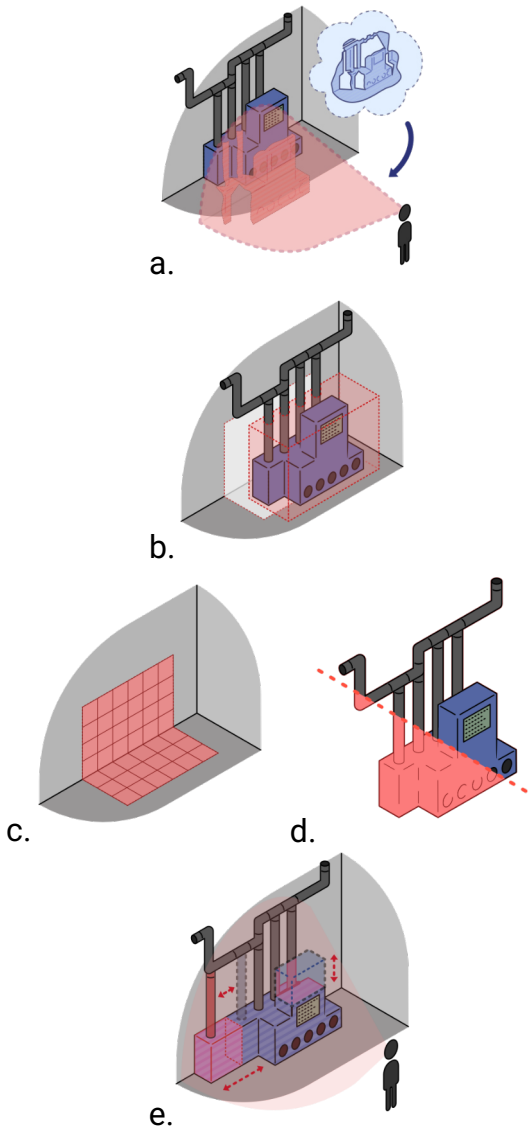


Figure 1.2: Overview of the proposed modular pipeline for transforming partial scans into dynamic reality models.

Chapter 2

Scene Dynamification Background and Terminology

2.1 Introduction

The previous chapter introduced the DRM as a representation that combines the geometric and visual fidelity of reality capture with the structured, interactive nature of object-based models, and identified the pipeline of stages required to produce it. Before presenting each stage in detail, this chapter establishes the conceptual and terminological foundation on which the remainder of this thesis rests.

Building on the representational definition of Chapter 1, we now define *scene dynamification* as the *process* of producing a DRM: transforming a static, unstructured 3D reconstruction into a structured collection of individually manipulable objects, embedded within a geometrically and visually complete environment. Rather than treating a scanned scene as a monolithic surface, dynamification introduces an explicit object-level decomposition that enables operations such as repositioning, replacement, scaling and removal, while preserving consistency with the captured geometry and appearance.

This chapter defines the scope and requirements of dynamification and positions it relative to existing representations. Section 2.2 reviews the landscape of 3D representations and discusses their suitability as inputs and outputs of the proposed pipeline. Section 2.3 formalises the requirements of a DRM in terms of its inputs, outputs and structural properties, including a typology of the

occlusion categories that the pipeline must address. Section 2.4 provides the conceptual description of the pipeline stages and explains the design rationale for treating scene and object completion in parallel. Section 2.5 surveys existing work that most directly addresses scene dynamification as a unified problem. Section 2.6 concludes with a summary of the scope and assumptions adopted throughout this thesis.

2.2 3D Representations

A prerequisite for understanding scene dynamification is a clear picture of the representational choices available for encoding 3D geometry and appearance. These choices have direct consequences for what operations are possible, what completion methods can be applied, and how the final dynamic scene can be consumed by downstream applications. Representations can be broadly divided into *explicit* and *implicit* categories, each with distinct trade-offs in terms of resolution, editability, storage efficiency and support for texture.

2.2.1 Explicit Representations

Explicit representations directly encode the geometry of a surface or volume as a discrete set of geometric primitives. They are generally well suited for visualization and downstream processing, as the geometry is directly accessible without the need for additional decoding steps.

Point clouds. A point cloud is an unordered set of 3D coordinates, typically accompanied by surface normals and colour values per point. Point clouds are the natural output of laser scanning and depth-sensing devices, and they form the primary raw input modality for the pipeline developed in this thesis. Their main advantages are simplicity, scalability and direct correspondence to physical measurements. However, they lack surface connectivity, which complicates downstream operations such as rendering, meshing and UV mapping. Colour is typically stored per point as an RGB triplet, either captured directly by the sensor or assigned through projection from a co-registered image [6]. The density and noise characteristics of a point cloud depend strongly on the sensing technology used, ranging from sub-millimetre precision in high-end terrestrial laser scanners to centimetre-level accuracy in mobile and XR-based devices [42, 21].

Meshes. A polygonal mesh represents a surface as a set of vertices connected by edges and faces, most commonly triangles. Meshes provide explicit surface connectivity and are the standard representation for rendering, simulation and geometric processing. They support a rich set of operations including Boolean combinations, normal computation and UV parameterisation for texture mapping.

Texture in meshes is typically stored as a 2D image, indexed via UV coordinates assigned to each vertex. Each face samples its colour from this texture image using barycentric interpolation of the UV coordinates of its three vertices [23]. This decoupling between geometry and appearance allows high-resolution textures to be stored efficiently, but introduces a non-trivial mapping between the 3D surface and the 2D texture space, which can complicate texture inpainting and completion [40].

Meshes obtained from indoor scans are typically the result of a reconstruction pipeline applied to point clouds or depth images, and may therefore contain artefacts such as holes, non-manifold geometry or incorrect connectivity [22]. Processing such meshes requires dedicated repair steps before further operations can be applied [3].

3D Gaussian splatting. A more recent explicit representation is 3D Gaussian splatting [25], in which the scene is represented as a set of anisotropic 3D Gaussians, each parameterised by a position, covariance, opacity and spherical harmonic coefficients encoding view-dependent colour. This representation has gained considerable attention for its ability to achieve real-time novel view synthesis at high visual fidelity. Each Gaussian implicitly encodes both geometry and appearance, analogous to coloured point clouds but with a richer per-element appearance model. While segmentation of Gaussian splat scenes into individual objects is an active area of research [2, 51], mature tools for object-level geometric editing, completion and part-aware deformation of Gaussian representations do not yet exist. Furthermore, the lack of an explicit, watertight surface definition makes completion and UV-based texture operations difficult to apply. For these reasons, 3D Gaussian splatting is not adopted as an intermediate or output representation in this thesis.

2.2.2 Implicit Representations

Implicit representations encode geometry indirectly, typically as a continuous scalar field over 3D space whose zero level set defines the object surface. They offer smooth, resolution-independent surface descriptions and are particularly

well suited as intermediate representations in learning-based completion and generation methods.

Signed Distance Functions. A SDF assigns to each point in space a scalar value representing the signed distance to the nearest surface, with negative values inside the object and positive values outside. The surface itself is recovered as the zero level set of this function. When discretised over a regular voxel grid, an SDF provides a fixed-size, differentiable representation that is compatible with convolutional neural network architectures [29, 35]. This makes SDFs a popular input and output format for object completion networks [12, 34, 15, 49]. However, conventional SDFs require the underlying geometry to be watertight, a condition rarely satisfied by raw scan data. UDFs, which encode only the absolute distance to the surface, provide a more general alternative applicable to open or incomplete meshes [43].

For the dynamification step, SDFs can be extended to CSDFs by additionally storing per-voxel colour information, enabling joint geometry and appearance manipulation [45, 31].

Parametric and semantic models. BIM models represent buildings as structured collections of parametric objects with associated semantic properties. Each element, such as a wall, door or piece of furniture, is described by its type, dimensions and relationships to other elements, rather than by explicit surface geometry. This structured representation supports efficient querying, simulation and modification, but typically lacks the geometric and visual detail of captured data [5]. As discussed in Chapter 1, the gap between BIM and reality capture representations motivates the development of the DRM.

Neural implicit representations. NeRFs [28] and related approaches represent scenes as learned continuous functions mapping 3D coordinates and viewing directions to colour and volume density. These representations achieve compelling novel view synthesis quality but are generally not editable at the object level, making them unsuitable as output representations for dynamification [47]. Recent methods attempt to decompose NeRF scenes into individual objects [2], but this remains an active research challenge and is not further pursued in this thesis.

2.2.3 Representational Choice in This Thesis

Given the above considerations, the pipeline developed in this thesis operates primarily with coloured point clouds and textured meshes as the main data types, with SDFs and UDFs as intermediate representations in completion stages. Point clouds serve as the natural output of the capture devices used and as the input to detection and completion modules [33]. Textured meshes serve as the output of the scene completion stage, providing a compact, renderable and editable surface representation [40]. SDFs are used internally within object completion networks, bridging the gap between the irregular scan data and the fixed-size inputs required by learning-based models [29, 13].

This choice reflects a deliberate balance between data fidelity, geometric expressiveness and compatibility with standard tools and processing pipelines [7]. BIM and neural representations are explicitly excluded from the pipeline output, as they either lack visual fidelity or do not support the object-level editing operations required for dynamification [5].

2.3 Dynamic Reality Model Requirements

Having established the representational landscape, we can now formalise what a DRM must provide and what inputs it operates on. This section defines the scope and structural requirements of the proposed representation.

2.3.1 Inputs

The pipeline takes as input heterogeneous, multi-source data captured from real indoor environments. This data is inherently incomplete, as no single scan captures a scene in its entirety. Two primary modalities are considered.

Point clouds. Coloured, unstructured point clouds form the geometric backbone of the input data. These may originate from high-end terrestrial laser scanners, which provide millimetre-level accuracy over large ranges, or from lower-end mobile and XR-based devices, which sacrifice accuracy for speed and accessibility [42]. As discussed in Chapter 1, no single scan captures a scene in its entirety: the point cloud covers only surfaces visible from the scanner positions, leaving occluded regions unobserved. Combining multiple scans from different positions increases coverage but introduces alignment challenges due to noise and drift [41].

Image data. Panoramic images or localised pinhole camera images provide complementary appearance information. Panoramic images, captured at the same positions as the scan data, enable dense colour assignment to the point cloud through angular projection. Localised images from mobile devices provide additional texture cues and can support 2D-based alignment and inpainting operations. The co-localisation of images within the same coordinate system as the point cloud is a prerequisite for cross-modal processing [7].

2.3.2 Output Requirements

The output of the pipeline is a DRM: a structured, complete and interactive representation of the indoor environment. This representation must satisfy the following requirements.

Geometric completeness. Both the structural elements of the scene and the individual objects within it must be geometrically complete. Occluded or unobserved regions must be reconstructed so that the resulting representation does not contain holes or missing surfaces. This applies at two levels: the background scene, consisting of walls, floors and ceilings, and the individual foreground objects, such as furniture or equipment.

It is useful to distinguish three categories of occlusion that the pipeline must address. *Sensor occlusion* refers to regions that fall outside the scanner's line of sight entirely, such as the back of a room not covered by any scan position. *Inter-object occlusion* refers to surfaces of one object that are hidden behind another, such as the rear face of a shelf visible only if the objects in front of it are removed. *Contact occlusion* refers to surfaces hidden because two objects are physically touching, such as the base of a chair leg resting on a floor or the back of a cabinet flush against a wall. Each category requires a different treatment: sensor occlusions can often be resolved by additional scan positions or scene-level inference [36], inter-object occlusions require object-level reasoning [43], and contact occlusions require knowledge of both the object geometry and the surrounding scene structure [44]. The occlusion analysis developed in Chapter 5 explicitly estimates all three types and passes this information to the scene and object completion stages.

Visual completeness. The completed geometry must be accompanied by plausible texture. Missing texture regions that arise from occlusions or incomplete image coverage must be inpainted in a manner that is consistent

with the surrounding appearance. The resulting textured representation must be visually coherent across the entire scene.

Object-level decomposition. Individual objects must be represented as separate entities within the scene, rather than as part of a monolithic surface. This decomposition is essential for enabling object-level manipulation. Each object must be isolatable from the scene without introducing inconsistencies in the surrounding environment.

Part awareness and scalability. To support realistic manipulation, completed objects must carry a notion of their internal part structure. This enables part-aware scaling operations, in which individual structural components of an object — such as the legs or seat of a chair, or the shelves of a bookcase — can be scaled independently or repeated, rather than applying a uniform deformation to the entire object. Uniform scaling of complex objects frequently produces geometrically implausible results by distorting repeating structural elements; part-aware scaling preserves the proportionality of individual components while adapting the overall dimensions [45, 48].

Part awareness in this pipeline is not a post-hoc annotation step but emerges naturally from the object completion process. The material segmentation performed during texture completion in Chapter 7 assigns discrete material indices to distinct surface regions of each object [43, 40]. These indices are propagated to the completed geometry and serve directly as the input to the convex decomposition and scaling zone definition in Chapter 8 [48]. This coupling ensures that the part structure used for scaling is grounded in the actual material composition of the completed object rather than in a generic, class-specific template.

Manipulability. Objects within the DRM must support a set of basic manipulation operations: repositioning within the scene, removal from the scene, and scaling. These operations must be applicable without re-running the full pipeline and without introducing visual artefacts in the surrounding scene.

2.4 Pipeline Overview

The requirements defined above are addressed by a modular pipeline, each stage of which is developed in detail in subsequent chapters. This section provides the conceptual description of each stage and its role within the overall workflow,

fulfilling the pointer made in Chapter 1. A key design decision running through the pipeline is the parallel treatment of scene and object completion, which is motivated here before the individual stages are described.

2.4.1 The Scene–Object Completion Duality

A central challenge in producing complete dynamic scenes is that scene-level and object-level completion are mutually dependent yet conventionally treated as separate problems. Scene-level completion methods [36, 19, 47] infer missing geometry by reasoning over the environment as a whole. This works well for regions that were simply outside the scanner's field of view, but fails at contact occlusions: surfaces hidden because two objects are physically touching cannot be distinguished from empty space by a global volumetric model, which has no concept of object identity or boundary.

Object-level completion methods [29, 13, 35, 53] operate on objects after they have been isolated from the scene. Isolation, however, strips away the environmental context that is most informative for completion. The floor surface constrains the bottom extent of a chair; the wall behind a cabinet defines its maximum depth; a neighbouring object provides an upper bound on where one object can end and another begin. A completion network that operates on a centred, decontextualised object mesh has none of this information available [44].

The proposed pipeline resolves this mutual dependency by decoupling detection from completion while keeping both coupled through a shared geometric analysis. Object detection produces bounding boxes that are refined using scene geometry — walls, floors and symmetry axes — before any completion is attempted [44, 38]. A voxel-based occlusion analysis then produces separate grids at the object level and scene level [46]. These grids encode which regions are genuinely missing versus which were simply not observed, and they are passed simultaneously to the scene completion stage, which uses them to avoid reconstructing geometry in regions occupied by detected objects, and to the object completion stage, which uses them to define the voxels that the completion network must predict [43]. The result is a pair of complementary reconstructions that are geometrically consistent with one another from the outset.

2.4.2 Pipeline Stages

Scene alignment and updating. Before any completion or decomposition can take place, multi-source and multi-temporal data must be brought into a common coordinate system. This stage handles the registration of point clouds

and images from heterogeneous devices and integrates incremental updates from lower-end sensors into existing high-quality datasets [42]. The output is a consistently georeferenced, multi-modal dataset representing the current state of the environment [7]. This stage is addressed in Chapter 4.

Object detection and occlusion analysis. The aligned point cloud is decomposed into individual objects using 3D detection methods [32]. Each detected object is associated with an oriented bounding box that is refined using symmetry estimation [38] and scene geometry constraints such as walls and floors. In parallel, a voxel-based occlusion analysis estimates the occluded regions of both individual objects and the full scene, distinguishing the three occlusion categories defined in Section 2.3.2 [46]. The resulting grids provide the shared geometric information that couples the scene and object completion stages. This stage is addressed in Chapter 5.

Scene completion. The structural elements of the environment, separated from the detected objects, are completed geometrically and texturally. Planar structures such as walls, floors and ceilings are reconstructed using plane detection and intersection-based geometry generation [40], guided by the scene-level occlusion grid to restrict reconstruction to genuinely missing regions and to avoid filling space occupied by detected objects. Texture is inpainted on the reconstructed geometry using the panoramic image data as reference [40, 39]. The output is a complete, textured mesh of the background scene. This stage is addressed in Chapter 6.

Object completion. Individual objects are completed independently, using both the partial point cloud and the co-registered image data as input [43]. The object-level occlusion grid defines which voxels the completion network must predict, while the refined bounding box — constrained by the scene planes — defines the physical extent within which the completion is permitted [44]. The completed objects are represented as textured meshes and carry part-index information derived from material segmentation [43], enabling the subsequent dynamification step. This stage is addressed in Chapter 7.

Scene dynamification. The completed objects are transformed into manipulable entities by applying part-aware scaling and decomposition [48]. Each object's internal structure is analysed and divided into approximately convex components that serve as the basis for scaling zones, using the material indices produced during object completion as structural priors [43]. The completed

scene and its objects are then assembled into a DRM in which each object can be independently repositioned, scaled or removed. This stage is addressed in Chapter 8.

2.5 Existing Approaches to Scene Dynamification

Although the individual components of the proposed pipeline, such as object detection [32], completion [29, 13] and inpainting [39, 40], have been studied extensively in isolation, the specific problem of transforming a partial scan into a fully dynamic and editable scene representation has received comparatively little attention [2].

The closest relevant body of work includes methods for indoor scene understanding that decompose a scan into structural and object components. Li et al. [27] segment indoor point clouds into planar structural elements and detected objects, reconstruct the structural geometry via plane fitting [26], and replace detected objects with matched CAD models retrieved from a database [37]. This approach produces a lightweight and semantically structured scene, but is limited to objects that have a close match in a predefined database and does not address the visual completeness of the reconstruction.

Methods based on neural implicit representations, such as NeRFiller [47] and related scene completion approaches [20, 11], use generative 2D inpainting to complete missing regions within neural radiance field representations. While these approaches can produce visually plausible completions, the resulting representations are not decomposed at the object level and do not support object-level manipulation or explicit geometric editing [36].

Gen3DSR [2] reconstructs a dynamic 3D scene from a single RGB image by separating objects from the scene using 2D segmentation masks and completing each independently using depth estimation and generative models. This is conceptually the closest existing approach to the pipeline proposed here, but it relies on a single image as input, limiting geometric accuracy, and does not address the alignment, updating or part-aware scaling stages that are central to this work.

More broadly, the field of semantic scene completion [36] aims to predict complete geometry and semantics from partial observations, but typically treats the scene as a monolithic volume and does not produce object-level decompositions suitable for interactive manipulation.

In summary, while components of the proposed pipeline exist in the literature, no existing work integrates alignment, object-level completion, scene completion

and part-aware dynamification into a unified framework operating on multi-source, multi-temporal scan data. This gap motivates the contributions of the subsequent chapters.

2.6 Scope and Assumptions

Having established the terminological and conceptual foundation, we conclude this chapter by summarising the scope and assumptions that govern the remainder of this thesis.

Scene type. The pipeline is designed for single-room indoor environments containing furniture or industrial objects. This constraint reflects the focus on densely furnished scenes where occlusions are significant but the scene structure remains largely planar. Outdoor environments, multi-room layouts and highly organic or curved architectural elements are outside the scope of this work.

Input modalities. The pipeline operates on coloured point clouds and panoramic or localised images. At least one high-quality baseline scan and one set of co-registered images are assumed to be available. The pipeline is designed to accommodate additional incremental observations from lower-end devices.

Object types. The pipeline targets geometrically distinct, rigid objects that are at least partially observed in the scan data. Deformable objects, transparent or highly reflective surfaces, and objects with no observable geometry are not addressed.

Output representation. The DRM is represented as a textured mesh for the background scene and as coloured SDFs or textured meshes for individual objects, with associated part indices. This representation is compatible with standard rendering and game engine pipelines.

Evaluation. Quantitative evaluation is performed primarily on the synthetic V-Scan dataset introduced in Chapter 3 [46], which provides ground-truth geometry for both the scene and individual objects. This synthetic dataset is a methodological necessity rather than a convenience: because real-world scans cannot provide ground truth for occluded regions — the surfaces that completion methods are specifically required to reconstruct — no real-world benchmark can directly assess completion quality in the regions that matter most [17, 9]. Real-world datasets are therefore used for qualitative evaluation and to assess generalisation beyond the synthetic domain.

These assumptions and constraints are revisited in Chapter 9, where the limitations of the proposed pipeline are discussed alongside directions for future

work.

Chapter 3

Data Acquisition

3.1 Introduction

The pipeline introduced in Chapter 2 requires data that satisfies several demanding conditions simultaneously. The input must consist of partially scanned indoor single-room scenes with accompanying colour information, captured from both industrial and residential environments. To support the alignment and updating stage (Chapter 4), multi-temporal and multi-sensory captures of the same environment are needed. The object detection stage (Chapter 5) requires ground-truth annotations for detectable objects, including bounding boxes. The scene completion stage (Chapter 6) requires ground-truth geometry of the empty scene, while the object completion stage (Chapter 7) requires both ground-truth complete object geometry and isolated partial scans of individual objects.

No single existing dataset satisfies all of these requirements. Publicly available scene-level datasets such as ScanNet [17] and Matterport3D [9] provide real scans with instance segmentation annotations, but lack ground-truth geometry for occluded regions. Synthetic scene-level datasets offer controlled conditions but typically lack the noise characteristics and occlusion patterns of real scanning devices. At the object level, large synthetic repositories such as ShapeNet [10] provide a wide variety of complete models, but these are not scanned data and often contain self-intersecting geometry. Real object-level datasets such as Redwood [14] offer high-quality reconstructions but cover a limited number of objects.

To address these gaps, this chapter describes the data acquisition strategy

adopted in this thesis. Three complementary data sources are combined. First, publicly available datasets are leveraged for specific pipeline stages where they provide sufficient coverage (Section 3.2). Second, real-world multi-temporal and multi-sensory captures are recorded at the university campus, providing heterogeneous data from different scanning technologies for the alignment and updating stage (Section 3.3). Third, a novel synthetic dataset generation framework is developed, consisting of a virtual scanner and a procedural room layout generator, to produce the V-Scan dataset. This dataset provides scene-level captures with and without objects, ground-truth geometry for all detectable objects along with their bounding boxes and isolated scans, and multi-temporal and multi-sensory variations, all under controlled conditions that replicate real-world scanning characteristics (Section 3.4). Section 3.5 presents a comparative overview of all datasets used in this thesis, and Section 3.6 concludes the chapter.

3.2 Existing Datasets

Several publicly available datasets have become standard benchmarks for 3D scene understanding and object reconstruction tasks. This section reviews the datasets used in this thesis and discusses their characteristics and limitations with respect to the requirements defined above.

3.2.1 Scene-Level Datasets

ScanNet. ScanNet [17] is an RGB-D video dataset containing 2.5 million views across more than 1500 scans, annotated with 3D camera poses, surface reconstructions, and instance-level semantic segmentations. Scenes are captured using commodity RGB-D sensors and reconstructed using BundleFusion. The dataset covers a wide variety of indoor environments including offices, bedrooms, kitchens, and living rooms. ScanNet provides per-vertex semantic and instance labels but does not include ground-truth geometry for occluded regions, limiting its use for quantitative evaluation of completion methods. The dataset is used in this thesis primarily for qualitative evaluation and to assess generalisation of detection methods.

ScanNet++. ScanNet++ [50] extends the ScanNet paradigm with significantly higher fidelity. The dataset currently contains over 1000 scenes, each captured with a high-end laser scanner at sub-millimetre resolution, registered 33-megapixel DSLR images, and commodity RGB-D streams from an iPhone. Scene reconstructions are annotated with an open vocabulary of semantics. The

combination of high-quality laser scans and commodity-level captures in the same scenes makes ScanNet++ particularly relevant for evaluating methods that operate on heterogeneous data. However, as with ScanNet, no ground-truth completion data is available.

Matterport3D. Matterport3D [9] provides 10,800 panoramic views from 194,400 RGB-D images across 90 building-scale scenes, captured with the Matterport Pro 3D Camera. The dataset includes textured surface reconstructions, globally aligned camera poses, and both 2D and 3D semantic segmentations at the instance level. Scenes span entire buildings, covering multiple rooms across several floors, and include a diverse set of residential environments. The building-scale coverage and panoramic image availability make Matterport3D well suited for evaluating scene-level methods that operate on panoramic imagery. The dataset is used in this thesis for qualitative evaluation of the scene completion and object completion stages.

ARKitScenes. ARKitScenes [4] provides a large collection of indoor scenes captured with Apple ARKit on iPad Pro devices equipped with LiDAR sensors. The dataset includes RGB-D video streams, camera poses, and 3D annotations for oriented bounding boxes of common indoor objects. ARKitScenes represents the class of commodity-level capture devices that the alignment stage of this pipeline must accommodate.

3.2.2 Object-Level Datasets

ShapeNet. ShapeNet [10] is a large-scale repository of 3D CAD models organised according to the WordNet taxonomy, containing over 51,000 unique models across 55 common object categories. The models are clean, watertight, and consistently oriented, making them well suited as training data for object completion networks. However, as synthetic CAD models, they differ substantially from scanned data in terms of surface noise, scan artefacts, and occlusion patterns. ShapeNet is used in this thesis as training data for the object completion networks described in Chapter 7.

Redwood. The Redwood 3D Scan Dataset [14] provides high-quality 3D reconstructions of real-world objects captured with consumer-grade RGB-D sensors. The dataset includes both individual object scans and full scene reconstructions. While the object reconstructions are detailed and realistic, the

dataset covers a relatively limited number of objects compared to synthetic alternatives, restricting its use as a comprehensive training set.

3.3 Real-World Data Capture

3.3.1 Introduction

The first set of manually scanned data used in this thesis consists of real-world captures performed at the KU Leuven Technology Campus in Ghent, Belgium. These captures serve two primary purposes: providing multi-temporal and multi-sensory data for evaluating the alignment and updating pipeline (Chapter 4), and providing real indoor scenes with diverse object configurations for qualitative evaluation of the detection and completion stages. The captures cover a range of industrial laboratory spaces and residential-style environments, using a combination of high-end terrestrial laser scanners and consumer-grade XR devices.

3.3.2 Sensing Technologies

The sensing technologies used in this thesis span a wide range of accuracy, density, and operational characteristics. Their properties directly influence the nature of the captured data and the challenges that each pipeline stage must address.

Terrestrial Laser Scanners. Static TLS systems such as the Leica ScanStation P30 provide the highest measurement accuracy, typically achieving single-point accuracies below 5 mm at ranges up to 50 m [6]. These systems capture structured point grids of 10 to 40 million points per setup, along with panoramic imagery. However, the static nature of the measurement introduces significant occlusions at ground level and behind objects, as only surfaces visible from the scanner position are recorded. Consecutive setups must be registered, typically through a combination of target-based and cloud-to-cloud methods, with sub-millimetre error propagation achievable across setups [22].

Indoor Mobile Mapping Systems. Indoor mobile mapping systems (iMMS) such as the NavVis VLX combine LiDAR sensors with camera-based SLAM algorithms to achieve continuous data acquisition while the operator moves through the environment. These systems produce unordered point clouds

with uneven point distributions, but their continuous trajectory results in significantly better spatial coverage than static scanners, particularly in cluttered environments. Global accuracies of LOA20 ($2\sigma \leq 0.05\,\mathrm{m}$) to LOA30 ($2\sigma \leq 0.015\,\mathrm{m}$) are achievable when supported by control measurements [22, 42]. Data acquisition is typically two to three times faster than with static TLS.

XR Devices. Extended Reality devices such as the Microsoft HoloLens 2 represent the lower end of the capture spectrum. These head-mounted devices are equipped with a 1 MP ToF sensor and four grey-scale tracking cameras, producing real-time polygonal meshes at a default spatial resolution of approximately $0.08\,\mathrm{m}^3$ [22]. While their accuracy is limited, with single-point accuracies of approximately 1 cm/m and a maximum useful range of 3–4 m, they offer unparalleled mobility and can capture regions inaccessible to larger systems. The accompanying RGB camera provides localised imagery that can be used for 2D-based alignment methods.

Smartphone LiDAR. Recent smartphones equipped with LiDAR sensors, such as iPhone models with the Apple LiDAR Scanner, provide another commodity-level capture modality. These devices produce dense point clouds at close range with moderate accuracy, and their widespread availability makes them relevant for incremental updating scenarios where dedicated scanning equipment is not practical.

3.3.3 Capture Methodology

Each sensing technology was deployed according to its intended use case. For TLS captures, one or two scanner setups were placed per room, positioning the scanner to maximise coverage of the structural elements while accepting the inherent ground-level occlusions caused by furniture and equipment. For iMMS captures, the operator walked through the environment following a natural trajectory, ensuring loop closures where possible to minimise drift accumulation. For XR device captures, the operator walked around the room attempting to capture all accessible surfaces, including regions underneath and behind furniture that are typically occluded in TLS data.

This deliberate combination of sensors and deployment strategies produces complementary datasets: TLS data provides high-accuracy structural geometry with systematic occlusions, iMMS data provides high-coverage reconstructions with moderate accuracy, and XR data provides fast, accessible captures with reduced accuracy but improved coverage in cluttered areas.

3.3.4 Captured Scenes

Multiple indoor environments were captured across the KU Leuven campus. These include industrial laboratory spaces used for concrete processing, road construction research, and building physics experiments, as well as a residential-style living lab environment. Each environment was captured at multiple time points and with multiple sensors, producing the multi-temporal and multi-sensory datasets required for the alignment experiments in Chapter 4. Between capture epochs, the environments were in active use, resulting in natural variations in object placement and configuration.

The captured data from these campaigns has been previously reported and analysed in [22, 42, 41], where the sensor capacity, alignment accuracy, and data processing characteristics were evaluated in detail.

3.3.5 Comparison of Sensing Technologies

The sensor comparison studies reported in [22] provide quantitative evidence for the complementary nature of the different capture technologies. The TLS achieves the highest point accuracy (LOA40) but the lowest spatial coverage (average 41% across structure classes), primarily due to the limited number of setups and the resulting systematic occlusions. The backpack-based iMMS (NavVis VLX) achieves the highest coverage (average 60%) while maintaining LOA20 to LOA30 accuracy, making it the most balanced system for general documentation. The HoloLens 2 achieves coverage comparable to the iMMS in accessible areas but is limited by its short range, resulting in poor coverage of ceilings, high walls, and distant surfaces. Its point accuracy is restricted to LOA10 to LOA20.

An important finding from [22] is that the differences in point cloud characteristics between sensors have a limited impact on downstream semantic segmentation performance. This sensor invariance opens the door to combining training data from multiple sources and to using synthetic data for training deep learning models, which directly motivates the synthetic dataset generation described in the following section.

3.4 Synthetic Data Creation

3.4.1 Introduction

While the real-world captures described in the previous section provide heterogeneous multi-sensory data for alignment evaluation, they cannot provide the ground-truth geometry required for quantitative evaluation of scene and object completion. The occluded surfaces that completion methods must reconstruct are by definition not observed in any real scan. This motivates the development of a synthetic dataset generation framework that produces partial scans paired with complete ground-truth geometry under controlled conditions that replicate real-world scanning characteristics.

The framework consists of two major components: a virtual scanner that simulates the acquisition process of real scanning devices (Section 3.4.3), and a procedural room layout generator that produces diverse furnished indoor environments (Section 3.4.4). Together, these components enable the automated generation of the V-Scan dataset (Section 3.4.6).

3.4.2 Related Work

Synthetic datasets have been widely used in 3D computer vision to overcome the limitations of real-world data collection. ShapeNet [10] provides a large collection of clean CAD models but does not simulate scanning artefacts. Structured3D [52] generates photorealistic indoor scenes with ground-truth annotations but does not simulate point cloud acquisition. Virtual scanning approaches have been explored in the context of autonomous driving, where LiDAR simulators generate realistic point clouds from virtual environments. In the indoor domain, our previous work [21] demonstrated that synthetic training data generated from virtual scanning can be combined with real data to improve semantic segmentation performance. However, existing virtual scanners typically focus on a single scanner type and do not provide the multi-sensory simulation required for this thesis.

3.4.3 Virtual Scanner

The virtual scanner is implemented as a Unity component that simulates the acquisition process of different scanning technologies within a 3D scene. The scanner operates from a fixed position within the scene and produces a coloured

point cloud along with a panoramic image, replicating the output of a terrestrial laser scanner or similar device.

Scanner Configuration

The scanner behaviour is controlled by a set of configurable parameters encapsulated in a `ScannerTypeScriptableObject`. These parameters define the characteristics of the simulated sensing device:

- **Scan density**: the angular spacing between consecutive rays, expressed as the point spacing in mm at 10 m distance. This parameter controls the total number of rays cast and thus the density of the resulting point cloud.
- **Scan range**: the maximum distance at which surfaces are detected, in metres.
- **Vertical scan range**: the total angular extent of the vertical field of view, in degrees. A value less than 360° simulates the blind spots typically found at the zenith and nadir of terrestrial laser scanners.
- **System noise**: the standard deviation of the range-independent measurement noise, in mm. This component models the intrinsic noise floor of the sensor electronics.
- **Distance noise**: the standard deviation of the range-dependent measurement noise, in mm per 10 m. This component models the increase in measurement uncertainty with distance.

To replicate the characteristics of real scanning devices, a set of scanner type presets was created based on manufacturer specifications. Table 3.1 summarises the parameters of the simulated scanner types used in this thesis.

Geometric Scanning

The scanning process operates by casting rays from the scanner position into the scene. The scan directions are computed to approximate the spherical scanning pattern of a terrestrial laser scanner. For a given angular density δ, the number of horizontal points per disc and the number of vertical discs are determined by:

$$n_{\text{horiz}} = \left\lceil \frac{2\pi}{\delta} \right\rceil, \qquad n_{\text{vert}} = \left\lceil \frac{\pi}{\delta} \right\rceil \tag{3.1}$$

Table 3.1: Scanner type presets used in the virtual scanning framework, based on manufacturer specifications of real devices.

Scanner Type	**Density** (mm/10m)	**Range** (m)	**Vert. Range** (°)	**Sys. Noise** (mm)	**Dist. Nois** (mm/10m
Leica P30	12.5	100	290	1.0	1.0
Leica BLK360 Dense	6.0	60	300	4.0	7.0
Leica BLK360 Fast	12.5	60	300	4.0	7.0
Matterport Pro2	25.0	9	260	3.0	5.0
Matterport Pro3	20.0	30	280	2.0	3.0
iPhone LiDAR	50.0	5	180	10.0	10.0
HoloLens 2	80.0	4	120	10.0	15.0

where $\delta = 2\sin(\text{density}/10000)$ converts the density parameter from mm at 10 m to radians. Rays outside the configured vertical scan range are discarded to simulate the blind zones of real scanners.

For computational efficiency, the raycasting is parallelised using Unity's `RaycastCommand` batch processing within the C# Job System. All rays are scheduled as a single batch and processed in parallel, significantly reducing computation time compared to sequential raycasting.

When a ray intersects a scene surface within the configured scan range, a scanned point is recorded with its 3D position, surface normal, and UV coordinates for subsequent colour assignment. If noise simulation is enabled, the measured distance is perturbed by Gaussian noise generated using the Box–Muller transform:

$$d' = d + \mathcal{N}\left(0,\ \sigma_{\text{sys}} \cdot 10^{-3} + \sigma_{\text{dist}} \cdot 10^{-4} \cdot d\right) \tag{3.2}$$

where d is the true distance, σ_{sys} is the system noise in mm, and σ_{dist} is the distance noise in mm/10m.

Image Capture and Colouring

Colour information is captured through a spherical image render using a dedicated camera component (`SphericalImageCapture`). The camera renders an equirectangular panoramic image from the scanner position, which serves both as the panoramic image output and as the source for point cloud colouring.

Each scanned point is assigned a colour by sampling the panoramic image at its corresponding UV coordinates. This sampling is parallelised using a `IJobParallelFor` implementation that maps each point's spherical UV coordinates to pixel indices in the rendered texture.

Object Isolation

A key feature of the virtual scanner for dataset generation is the ability to isolate individual objects from the scene scan. Each object in the scene is annotated with a `CaptureObject` component that defines its bounding box and provides methods for isolating the points that fall within its volume from the full scene point cloud. The isolation is performed by filtering the scanned points against the object's oriented bounding box.

For each isolated object, the framework exports the partial point cloud (the scanned points within the bounding box), the ground-truth mesh (the complete mesh geometry of the object), and the oriented bounding box parameters. Optionally, a per-object voxel occupancy grid can be computed to characterise the occlusion pattern within the object's bounding box.

Export Format

The virtual scanner exports point clouds in a text-based format compatible with CloudCompare, with each line containing the position, normal, and colour of a scanned point (nine values: x, y, z, n_x, n_y, n_z, r, g, b). A header block records the capture timestamp and the scanner's transformation matrix. Ground-truth meshes are exported in PLY format, and bounding boxes and scan parameters are serialised as JSON files.

3.4.4 Procedural Room Layout Generator

The procedural room generator produces randomised indoor environments consisting of a rectangular room structure populated with furniture objects. The generator is implemented as a Unity component (`ProceduralRoomGenerator`) that creates rooms from a library of modular prefab elements defined in a `PrefabListScriptableObject`.

Room Structure Generation

The room structure is generated on a grid-based system. The floor dimensions are randomly sampled within configurable bounds, and the floor, walls, and ceiling are assembled from prefab tiles placed on the grid. During wall generation, each wall segment has a configurable probability of being replaced by a door or window prefab, introducing structural variation across generated rooms. A door is always placed at one wall position to ensure accessibility.

Furniture Placement

Furniture placement is divided into two categories: wall-mounted furniture and free-standing furniture. Wall-mounted furniture (such as shelves, paintings, or cabinets) is placed against available wall segments with a configurable probability. Free-standing furniture is placed within the interior of the room using a rejection sampling strategy. A placement zone is defined by insetting the room boundary by one grid cell, and objects are placed at random positions and orientations (quantised to 90° increments) within this zone. Each candidate placement is checked against all previously placed objects using axis-aligned bounding box intersection tests, and rejected if an overlap is detected. The maximum number of furniture items is controlled by a configurable density parameter that scales with the floor area.

The room structure and furniture are maintained as separate hierarchies, allowing the furniture to be regenerated independently of the room structure. This separation enables the creation of multi-temporal variations of the same room with different furniture configurations, which is essential for generating the scan variations described below.

Seeded Randomisation

All random decisions in the generation process are controlled by a single seed value, ensuring full reproducibility. When a fixed seed is specified, the same room layout and furniture configuration are produced deterministically. When no fixed seed is provided, the current system time is used, producing a unique configuration for each generation.

3.4.5 Scan Management and Dataset Generation

The virtual scanner and procedural room generator are coordinated by a `ScanManager` component that orchestrates the automated generation of complete dataset entries. For each generated room, the scan manager executes the following sequence:

1. **Room generation**: A new room is procedurally generated, and all detectable objects are identified within the scan volume.
2. **Full scene scan**: The scene is scanned with objects present, producing the main point cloud and panoramic image. Ground-truth meshes and bounding boxes are exported for all detected objects.
3. **Object isolation**: Each detected object's points are isolated from the full scan, producing partial point clouds per object. Optionally, per-object occlusion voxel grids are computed.
4. **Empty scene scan**: All detected objects are deactivated, and the scene is scanned again to produce the ground-truth empty scene point cloud.
5. **Scan variations**: Additional scans are generated with configurable variations, including changes in furniture layout (by regenerating only the furniture while keeping the room structure), changes in scanner type (by switching to a different scanner preset), and changes in scanner position (by randomising the scanner location within the room volume).

Each variation is stored alongside the main scan, together with its scan parameters, enabling controlled experiments on the impact of individual factors such as scanner type, scanner position, and temporal object changes. The variation system supports an arbitrary number of variation passes per room, with each pass independently configurable.

3.4.6 The V-Scan Dataset

Using the framework described above, the V-Scan dataset was generated to serve as the primary evaluation benchmark for the scene and object completion stages of this thesis. The dataset is designed to provide the combination of properties that no existing dataset offers: scene-level partial scans paired with complete ground-truth geometry, multi-temporal and multi-sensory variations, and per-object ground truth including isolated partial scans, complete meshes, and oriented bounding boxes.

Dataset Composition

The V-Scan dataset contains 200 scene folders generated from 5 unique procedural room layouts (scene names), each scanned with multiple scanner type presets and variation passes. Table 3.2 summarises the key statistics of the dataset.

Table 3.2: Summary statistics of the V-Scan dataset.

Metric	**Value**
Total scene folders	200
Unique scene names	5
Total objects	3,100
Average objects per scene	15.5
Scanner types used	7

The point counts vary significantly across scanner types, reflecting the differences in scan density and range. Table 3.3 provides representative point counts for a single scene scanned with different virtual scanner presets.

Table 3.3: Representative point counts for a single scene (Industrial_1) across different scanner types.

Scanner Type	**Points**
Leica P30	2,529,458
Leica BLK360 Dense	∼5,000,000
Leica BLK360 Fast	∼1,500,000
Matterport Pro2	∼200,000
Matterport Pro3	∼400,000
iPhone LiDAR	∼50,000
HoloLens 2	∼20,000

Per-Scene Outputs

For each scene folder, the following outputs are available:

- **Main scan**: coloured point cloud of the furnished scene with the main scanner type.
- **Panoramic image**: equirectangular panoramic image from the scanner position.

- **Scan parameters**: JSON file recording the scanner type configuration used.
- **Room parameters**: JSON file recording the procedural generation parameters.
- **Object bounding boxes**: JSON file containing oriented bounding boxes for all detected objects.
- **Per-object data**: for each object, the isolated partial point cloud, the ground-truth complete mesh (PLY), and the individual bounding box.
- **Empty scene scan**: coloured point cloud and panoramic image of the scene with all objects removed.
- **Scan variations**: additional point clouds and panoramic images for each configured variation, along with variation parameters.

This comprehensive set of outputs enables evaluation at multiple levels: scene-level completion quality can be assessed by comparing the reconstructed empty scene against the ground-truth empty scan, object-level completion quality can be assessed by comparing completed objects against the ground-truth meshes, and detection quality can be assessed by comparing detected bounding boxes against the ground-truth annotations.

3.5 Dataset Comparison and Overview

Table 3.4 provides a comparative overview of all datasets used in this thesis. The comparison highlights the complementary nature of the data sources: real datasets provide realistic scanning characteristics but lack ground-truth completion data, while the synthetic V-Scan dataset provides comprehensive ground truth at the cost of reduced visual realism. By combining these sources, each pipeline stage can be evaluated using the most appropriate data.

The campus captures provide the multi-temporal and multi-sensory data required for the alignment experiments but lack ground-truth completion data. The public scene-level datasets provide large-scale, diverse real-world scenes for qualitative evaluation and generalisation assessment. ShapeNet and Redwood provide object-level data for training completion networks. The V-Scan dataset uniquely combines scene-level and object-level ground truth with multi-temporal and multi-sensory variations, enabling quantitative evaluation of the complete pipeline under cont rolled conditions.

Table 3.4: Comparative overview of datasets used in this thesis. GT = ground truth available, MT = multi-temporal captures, MS = multiple scanner types.

Dataset	Type	GT Compl.	MT	MS	# Scenes	Scene Types	Used For
ScanNet [17]	Real	×	×	×	1,513	Residential	Qual. eval.
ScanNet++ [50]	Real	×	×	✓	1,000+	Mixed	Qual. eval.
Matterport3D [9]	Real	×	×	×	90	Residential	Qual. eval.
ARKitScenes [?]	Real	×	×	×	5,047	Mixed	Detection
ShapeNet [10]	Synthetic	✓	–	–	–	Objects only	Training
Redwood [14]	Real	✓	×	×	–	Objects only	Training
Campus captures	Real	×	✓	✓	10+	Industrial	Alignment
V-Scan (ours)	Synthetic	✓	✓	✓	200	Mixed	Quant. eval.

3.6 Conclusion

This chapter presented the data acquisition strategy adopted in this thesis, addressing the diverse requirements of the proposed pipeline through a combination of three complementary data sources. Publicly available datasets provide large-scale, diverse real-world scenes for training, qualitative evaluation, and generalisation assessment. Real-world multi-temporal and multi-sensory captures from the university campus provide heterogeneous data for evaluating the alignment and updating pipeline. The novel V-Scan synthetic dataset, generated through a virtual scanning and procedural room generation framework, provides the comprehensive ground-truth data required for quantitative evaluation of scene and object completion methods.

The virtual scanning framework introduced in this chapter simulates the acquisition characteristics of seven different scanner types, ranging from high-end terrestrial laser scanners to consumer-grade XR devices, by configuring scan density, range, field of view, and noise parameters. The procedural room generator produces diverse furnished indoor environments with controllable complexity. Together, these components enable the automated generation of large-scale datasets with complete ground-truth geometry at both the scene and object level, multi-temporal variations through furniture regeneration, and multi-sensory variations through scanner type switching.

The V-Scan dataset, containing 200 scene folders with 3,100 total objects, serves as the primary benchmark for the quantitative evaluation conducted in the subsequent chapters. Its combination of scene-level and object-level ground truth, multi-temporal captures, and multi-sensory variations addresses the evaluation gap identified in Chapter 1, where the absence of ground truth for occluded regions was identified as a fundamental barrier to the quantitative assessment of completion methods.

Chapter 4

Alignment And Updating

Chapter 5

Scene Segmentation

5.1 Introduction

5.2 Related work

5.3 Methodology

5.3.1 Object detection

5.3.2 Environment segmentation

5.4 Experiments

5.5 Conclusion

Chapter 6

Scene Completion

6.1 Introduction

6.2 Related work

6.3 Methodology

6.3.1 Geometry reconstruction

6.3.2 Color inpainting

6.4 Experiments

6.5 Conclusion

Chapter 7

Object Completion

7.1 Introduction

7.2 Related work

7.3 Methodology

7.3.1 Geometry generation

7.3.2 Texture generation

7.4 Experiments

7.5 Conclusion

Chapter 8

Scene Dynamification

8.1 Introduction

8.2 Related work

8.3 Object Dynamification

8.3.1 Part segmentation

8.3.2 Scaling zone definition

8.3.3 Value interpolation

8.3.4 Value repetition

8.4 Scene Reconstruction

8.5 Experiments

8.6 Conclusion

Chapter 9

Conclusion and future works

9.1 Conclusion

9.2 Future works

Chapter 10

Valorisation

10.1 Introduction

10.2 Proof-of-concept applications

In this section, two tools developed up to the proof-of-concept stage are presented.

10.2.1 Digital upcycling

10.2.2 Furniture Removal

10.3 Valorization potential and market position

10.3.1 Time savings analysis

10.4 Conclusion

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Statement on the use of Generative AI

I did use the generative AI assistance tools ... Generative AI was used for generating programming code.
The text, code, and images in this thesis are my own (unless otherwise specified). Generative AI has only been used in accordance with the KU Leuven guidelines and appropriate references have been added. I have reviewed and edited the content as needed and I take full responsibility for the content of the thesis.

Curriculum Vitae

Jelle Vermandere was born on June 26, 1995. In 2020, he obtained his master's degree in Architectural Engineering at UGent, graduating with distinction. His thesis focussed on creating a website and database for interactive viewing of 3D models withing exhibitions.

After his studies, Jelle was a self-employed freelancer, creating interactive architectural renderings and video games for various clients.

In July 2021, he started his research at the Faculty of engineering technology at the KU Leuven under the supervision of Prof. Vergauwen and Prof. Bassier. A year later, he secured an FWO strategic basic research grand, allowing him to focus his main research which centers around innovating the built environment through Reality Modelling. The goal of this research is to create dynamic and complete 3D models of indoor environments captured using 3D scan-devices.

Jelle has a wide research interest, focussing around the combination of real and digital techniques. Outside his research, his hobbies led him to mastering the full real-to-digital-to-real workflow using laser scanning, 3D modeling and 3D printing, he can create almost anything. He documents his projects on his YouTube channel where he created a community of over 90.000 subscribes accumulating over 7 million views.

Jelle also likes to practice a wide range of sports including: running, biking, swimming and climbing. After all the hours behind his computer, the physical exercise is a very welcome distraction.

Publications

Journal Articles

1. Vermandere, J., Bassier, M., & Vergauwen, M. (2022). *From Partial Scans to Editable Scenes: Hybrid Object and Scene Completion for Dynamic 3D Environments.* **Automation in Construction**, 14(11), 2680. DOI: 10.3390/rs14112680

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